

Pranav Harwadekar

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EDUCATION

Texas A&M University

B.S. in Computer Science, Minor in AI and Statistics

GPA: 3.93/4.0

August 2023 – May 2027

President's Endowed Scholar

Engineering Honors Student

TECHNICAL SKILLS

Languages/Frameworks: Python, C++, Swift, Java, IBM Certified (React Flask NodeJS, ExpressJS), Express, HTML, CSS, JavaScript, TypeScript, MongoDB, Unix/Linux, Firebase, Git, OpenCV, NLTK, NumPy, Pandas, Seaborn, Matplotlib, SciPy, Selenium, AWS, Vercel, AutoCAD, Fusion360

Courses: Data Structures/Algorithms, OOP, Discrete Math, C Structured Programming, Computational Data Science, IT Practicum (Firebase)

EXPERIENCE

Software Engineering Intern

May 2026 – August 2026

Bumble Inc.

- Developing **LLM-powered agents** integrated with Vertex AI, DBT, Airflow, and Snowflake to automate feature store validation and machine learning pipeline workflows across **40+ TB of production data**.
- Designing **agent evaluation, testing, and guardrail frameworks** to validate feature definitions, data lineage, and YAML-based ML artifacts, improving reliability of production ML infrastructure.
- Collaborating with Recommendation Systems and Trust & Safety ML teams to build **scalable MLOps solutions** that accelerate feature engineering, workflow orchestration, and model deployment.

Software Engineer Intern

June 2025 – August 2025

McKesson Corporation

- Designed and implemented reorder point algorithms and drug dispense forecasting algorithms using **Apache Spark (PySpark)** and **MLflow** on **Databricks** to optimize drug demand prediction and reorder timing.
- Leveraged **spatial clustering** and seasonal drug patterns to enhance prediction performance across retail and health system chains.
- Enhanced model performance by automating feature engineering and applying quantile-based forecasting to minimize inventory inefficiencies.

Software Engineer / Autonomous Navigation Engineer Lead

September 2024 – Present

Karura USA

- Developing autonomous navigation algorithms in **C++/Python** for URC competition using ROS-based Nav2 and **Stereo SLAM** with RealSense D435 cameras, optimizing **GPS waypoint** following and **sensor fusion (EKF)** for precise localization across varied terrains.
- Simulating navigation and terrain algorithms in **Gazebo** and implementing **image detection** for ArUco markers.
- Boosted **AI navigation accuracy by 25%** through sensor data analysis and algorithm tuning with offshore Japanese team using Python and ROS.

Data Science Project Lead

September 2024 – December 2024

American Airlines x Aggie Data Science Club

- Developed **AI predictive models for baggage arrival timing** on over **3 million flights** from a Snowflake's Data Cloud, utilizing **decision trees** to capture complex relationships within flight operations.
- Analyzed model outputs to optimize ground operations, reducing passenger wait times to under 15 minutes for American Airlines.

Experience with Extern Inc

Beats by Dre (Apple) - Consumer Insights Data Analyst

June 2024 – August 2024

- Conducted sentiment analysis on **1,200 Amazon reviews** across **10 brands of headphones** using **manual and API** scraping.
- Chosen **among over 110** externs to present with the **Head of Consumer Insights at Beats**.
- Performed EDA analysis with Pandas, SciPy, and Seaborn and advanced NLP analysis with **Gemini API** for feature specific insights.

Webacy Inc - Web3 Security Data Analyst

May 2024 – June 2024

- Conducted rigorous **Exploratory Data Analysis** on smart contract risk tags analyzing over **965** lines of data across **270 contracts** from **3** distinct databases and presented visualization data from **SQL, Excel, Tableau**.
- Profiled risks into 1 of 30 bins with **unsupervised ML** techniques (hierarchical clustering and Jaccard distance) for **cluster analysis**.

Machine Learning Researcher

September 2023 – December 2023

Texas A&M University - Debakey Executive Research Program

- Implemented **hyperspectral algorithms** in **C++**, including Jeffries Matusita Distance (**JM**), Spectral Angle Mapper (**SAM**), to enable precise spectral signature identification from complex hyperspectral data. Used **WSL and Docker** for cross-platform development.
- Created a C++ GUI using Glade** and used **CUDA for parallel computing**, cutting hyperspectral data processing time by 40%.

PROJECTS

Autonomous Vehicle Navigation Simulation

September 2024 – Present

- Developing and integrating advanced sensor systems and navigation algorithms using **CARLA Simulator**, enhancing autonomous vehicle decision-making through machine learning (**TensorFlow Graphs/PyTorch Geometric**)
- Utilizing **GitHub** for seamless collaboration and version control and **Jupyter Notebook/Google Colab** for the DS/ML scripts.

AI for Justice – Criminality (3rd place overall round TIDAL Hackathon) ([Github](#))

October 2024 – October 2024

- Developed a real-time expression abnormality detector using **React** based front-end with **Flask and AWS** (S3 and Transcribe) for backend
- Integrated machine learning algorithms with **OpenCV** to analyze speech patterns and eye movement, detecting inconsistencies indicative of deception, while comparing the results to **NLP-extracted** testimony to assess credibility

Club3 Study Buddy Application Developer

July 2024 – Present

- Developing a collaborative study app with **Maps API** and **Socket Programming**, to connect users with a real-time group study session.
- Utilized **MongoDB** for user profile storage, **Clerk** for authentication, and **Google Analytics** to track user behavior and optimize app.